

Coarse-Grain Perturbation Robustness of CNN and Attention-Based Classifiers under THz-Like Low-Resolution Imaging

Itamar Bahat

ibahat90@gmail.com

Object Classification in Low-Resolution THz Imagery project (p-2026-061)

Pipeline v2 noise-fix campaign — branch `noise_fix`, closed 2026-05-23

Abstract—We evaluate the robustness of three deep-learning classifiers — ResNet50 [1], DenseNet121 [2], and TransNeXt-tiny [3] — under severe visual degradation that simulates Terahertz (THz) imaging conditions: low resolution, Gaussian blur, additive Gaussian noise, salt-and-pepper noise, and desaturation. A 186-cell experimental campaign at uniform 224×224 input resolution covers a five-level severity curve over two datasets (CIFAR-10 [4] and MNIST [5]) in three phases: clean baselines (6 cells), combined degradation (30 cells), and single-axis isolation (150 cells). We present three contributions. *First*, we identify and correct a bug in the degradation pipeline that caused additive Gaussian noise and salt-and-pepper to be drawn *after* the bicubic upsample to 224, where each stochastic sample landed on a single output pixel and was averaged out by the bicubic kernel. Our v2 pipeline draws both perturbations at `low_res` and bicubically spreads each sample across a wider footprint, producing the sensor-realistic coarse-grain structure a true low-resolution detector would record (lag-1 spatial autocorrelation of the residual jumps from ~ 0 to 0.993 at `low_res=8`). *Second*, under the corrected pipeline we identify a universal 3×3 -downsample bottleneck: at `low_res=3`, all three architectures collapse to within ± 3 pp on CIFAR-10 (0.43–0.46) and ± 1 pp on MNIST (0.44–0.45), regardless of whether the backbone is convolutional or attention-based. The pretrain receptive-field hierarchy fails uniformly when sub-class geometric structure is destroyed. *Third*, we establish a 60-cell two-axis empirical result that coarse-grain stochastic perturbations selectively destroy texture-dependent recognition (CIFAR-10) while preserving geometry-dominant recognition (MNIST). TransNeXt-tiny retains a single-digit-to-low-double-digit robustness gap over the CNN backbones on every CIFAR-10 perturbation axis at $L = 5$, but the absolute floor is much lower than the pre-fix numbers suggested: `val_acc` drops to 0.80 on noise (was 0.96 under the bug) and 0.94 on salt-and-pepper (was 0.96). The advantage disappears at $L = 5$ resolution. Attention softens but does not reverse the CIFAR-10 texture-statistic collapse. An appendix (§A) reports Phase D, a 90-cell regularization sweep that quantifies the residual recoverable accuracy: T3 (dropout + mixup + cutmix) yields a mean $+0.77$ pp recovery, confirming the information-bottleneck interpretation of the v2 collapse.

Index Terms—Image classification; Terahertz imaging; low-resolution imagery; visual degradation; robustness; vision transformers; CNN; deep learning; TransNeXt; DenseNet; ResNet; CIFAR-10; MNIST.

I. INTRODUCTION

Terahertz (THz) imaging is a non-ionising, sub-millimetre-wave sensing modality used in security screening, materials inspection, and biomedical contexts. Compared with optical

imaging it produces fundamentally low-resolution outputs: typical operational sensors deliver imagery on the order of ~ 3 –32 pixels per dimension before any super-resolution post-processing. The corresponding classification problem — recognising object categories under severely information-limited inputs — is therefore distinct from the more heavily studied natural-image classification regime [6] in two important ways. First, the input resolution is far below the $\sim 224 \times 224$ to which ImageNet-pretrained backbones are anchored, and second, the dominant nuisance variables are coarse-grain (sensor-pixel or larger) rather than fine-grain (photographic) perturbations.

In this work we benchmark three representative deep-learning backbones under a controlled, parametric THz-like degradation pipeline. We deliberately avoid the use of real THz imagery: real THz datasets are scarce, application-specific, and confound classifier behaviour with calibration, sensor-noise floor, and bandwidth artefacts that vary by instrument. Instead we degrade two widely-used clean datasets (CIFAR-10 [4] and MNIST [5]) through five orthogonal axes spanning the physical mechanisms that a real low-resolution detector imposes: spatial downsampling (*resolution*), additive Gaussian noise (*noise*), Gaussian smoothing (*blur*), additive salt-and-pepper spikes (*salt_pepper*), and luminance–chrominance attenuation (*saturation*). The five-level severity curve is calibrated to sweep from a mild near-clean regime ($L = 1$) to a 3×3 native input regime ($L = 5$) that mimics current consumer-grade THz scanners.

We evaluate three backbones chosen to cover the dominant architectural families: ResNet50 [1] (residual CNN, ~ 25 M parameters), DenseNet121 [2] (densely-connected CNN, ~ 8 M parameters), and TransNeXt-tiny [3] (aggregated-attention vision transformer, ~ 28 M parameters; we use the tiny variant for capacity parity with ResNet50). All three load ImageNet-pretrained weights and are fully fine-tuned with a uniform protocol at 224×224 input resolution. The fair-comparison invariant is therefore *same tensor shape, same training protocol, same degradation pipeline*; observed performance differences are attributable to architecture rather than to bookkeeping.

Beyond simply reporting headline numbers, we draw attention to a methodological correction that we believe is important for any future work in this area. The first version of our degradation pipeline (v1) applied additive Gaussian noise and

salt-and-pepper *after* the bicubic upsample to 224×224 — each `torch.randn` sample landed on a single output pixel and was effectively averaged out by the bicubic kernel that produced its neighbours. The result, at first glance plausible, was unrealistically high `val_acc` on the noise and salt-and-pepper axes: under `v1`, TransNeXt-tiny appeared to hold ≥ 0.95 on CIFAR-10 noise across $L = 1 \rightarrow 5$, even at `low_res=3`. We identified this as a software bug, not a property of the data: a true low-resolution sensor records noise at the sensor-pixel scale, not at the upsampled-pixel scale. Our `v2` pipeline draws the perturbations *at* `low_res` and lets the bicubic upsample spread each draw across a wider footprint at 224. The lag-1 spatial autocorrelation of the residual jumps from ~ 0 under `v1` to 0.993 under `v2` at `low_res=8`. We re-ran the 90 affected cells under `v2` (the 30 Phase B combined cells, the 30 Phase C noise cells, and the 30 Phase C salt_pepper cells) and present those `v2` numbers throughout. The 96 unaffected cells (Phase A clean plus Phase C resolution / blur / saturation) are byte-identical to `v1`.

A. Related Work

The robustness of CNN and attention-based image classifiers under visual perturbations has been studied along several distinct axes that are worth disentangling. Adversarial-robustness work studies imperceptible additive perturbations crafted to maximise classifier loss; that regime is fundamentally different from the sensor-realistic coarse-grain perturbations we study, where every perturbed pixel is also visibly perturbed to a human observer. Natural-image-corruption benchmarks (CIFAR-C, ImageNet-C, and their derivatives) characterise the per-class accuracy under a broad menu of distortions and have shown that attention-based architectures generally outperform CNNs on out-of-distribution robustness, but these benchmarks typically perturb at the native input resolution and do not isolate a low-resolution downsample axis separate from the perturbation axes. The TransNeXt [3] paper itself reports state-of-the-art ImageNet accuracy at every parameter budget and attributes the gain to aggregated foveal attention; we evaluate the `tiny` variant for parameter-budget parity with ResNet50, which makes the cross-architecture comparison here a cleaner test of attention-vs-convolution under degradation than would be obtained by comparing model sizes within the same family. The DenseNet121 [2] backbone is included for its dense-connection topology, which differs structurally from both the residual-block ResNet50 [1] and the patch-attention TransNeXt; the three together span the representative architecture families.

Within the THz-imaging literature, the dominant deep-learning applications are concealed-object screening, materials inspection, and biomedical micro-imaging, and the prevailing approach is to super-resolve the THz input to a CNN-compatible resolution before classification. Our work is complementary: rather than recovering information that the sensor has discarded, we ask how a high-quality classifier performs when the recovered input is itself coarse-grained. The five-axis severity curve in Table II sweeps the range from a ~ 18 -pixel “mild” downsampling regime (similar to many THz

TABLE I: Backbones evaluated in this study.

Model	Family	Params	Pretrain
ResNet50 [1]	residual CNN	$\sim 25\text{M}$	ImageNet-1K
DenseNet121 [2]	dense CNN	$\sim 8\text{M}$	ImageNet-1K
TransNeXt-tiny [3]	ViT (attention)	$\sim 28\text{M}$	ImageNet-1K

reconstruction benchmarks after super-resolution) down to a 3×3 “extreme” regime that matches a raw THz scanner before any reconstruction. The $L1 \rightarrow L5$ trajectory therefore brackets the operating range of plausible real-world THz classification pipelines.

Roadmap: Section II describes the degradation pipeline, including the `v1` $\rightarrow$ `v2` correction; Section III describes the 186-cell matrix, the Optuna [7] pre-tuning of hyperparameters at the moderate $L = 3$ severity, and the PyTorch Lightning [8] training engine; Section IV presents the `v2` numerical results across all three phases; Section V interprets the headline findings and contrasts them with the `v1` numbers; and Section VI discusses limitations and future directions.

II. METHODOLOGY

A. Datasets

We use CIFAR-10 [4] as a *texture-dominant* benchmark: 10 natural-image object classes at native 32×32 RGB with 10K training and 5K validation samples drawn from the standard torchvision split. MNIST [5] serves as a *geometry-dominant* contrast: 10 hand-written digit classes at native 28×28 grayscale (broadcast to 3 channels), with the same 10K/5K split. Both datasets are upsampled by the degradation pipeline to 224×224 before reaching any backbone, so input tensor shape is identical across all six (model, dataset) pairs.

The two datasets are chosen for contrast in the dominant discriminative feature. CIFAR-10’s per-class signal is carried by texture statistics that are spatially distributed across the image (e.g. feather patterns, fur, wing structure), while MNIST’s per-class signal is carried by the global geometric shape of thick digit strokes that are robust to local perturbations. We expect coarse-grain perturbations to bite sharply on the first and weakly on the second; the 60-cell two-axis result in Section IV confirms this directly.

B. Architectures

Table I summarises the three backbones. We chose them to span the three dominant families of supervised-pretrained image classifiers as of the current literature: residual CNNs, densely-connected CNNs, and aggregated-attention vision transformers.

Both CNNs are fine-tuned end-to-end with differential learning rates (head 10^{-3} , backbone 10^{-4}) following the TResNet [9] convention. TransNeXt-tiny is fine-tuned end-to-end with the paper-prescribed differential learning rates (head 5×10^{-4} , backbone 5×10^{-5}) per [3] §A.3. All three backbones load their official ImageNet-pretrained checkpoints [6] before training.

We deliberately avoid heavier TransNeXt variants (`small`, `base`) for capacity parity with ResNet50. The `tiny` variant

has approximately the same parameter budget ($\sim 28\text{M}$ vs $\sim 25\text{M}$); the alternative (`small` at $\sim 50\text{M}$) would confound the architecture comparison with parameter-count effects.

C. Degradation Pipeline

The degradation pipeline applies five orthogonal perturbation axes in the following order. Saturation is applied first, at native resolution, so that any downstream noise stays colour-correct. The image is then downsampled (bilinear) to the configured `low_res`; this is the sensor-pixel scale. Additive Gaussian noise and salt-and-pepper are then drawn at `low_res`. The image is upsampled (bicubic) to the model-input resolution of 224×224 . Gaussian blur is then applied with kernels calibrated for the 224×224 target. Finally, the image is clamped to $[0, 1]$ and normalised with the ImageNet channel statistics:

$$\begin{aligned} \mathbf{x}_{\text{sat}} &= (1 - s) \text{gray}(\mathbf{x}) + s \mathbf{x}, \\ \mathbf{x}_{\text{low}} &= \text{down}_{\text{bilin}}(\mathbf{x}_{\text{sat}}, r), \\ \mathbf{x}_{\text{nse}} &= \mathbf{x}_{\text{low}} + \sigma \mathcal{N}(\mathbf{0}, \mathbf{I}), \\ \mathbf{x}_{\text{sp}} &= \text{paint}(\mathbf{x}_{\text{nse}}, p_{\text{sp}}), \\ \mathbf{x}_{\text{up}} &= \text{up}_{\text{bicub}}(\mathbf{x}_{\text{sp}}, 224), \\ \mathbf{x}_{\text{out}} &= \text{blur}_{\text{Gaussian}}(\mathbf{x}_{\text{up}}, k, \sigma_b). \end{aligned} \quad (1)$$

The configurable parameters are $\{s, r, \sigma, p_{\text{sp}}, k, \sigma_b\}$ for `{saturation, low_res, noise_std, salt_pepper, blur_kernel, blur_sigma}`.

Pipeline v1→v2 correction: The fix is gated by a single integer constant `PIPELINE_VERSION=2` (defined in `src/data/degradation_levels.py` and stamped into each cell’s `metrics.json`) so downstream tooling can distinguish v1 and v2 cells programmatically. The pre-fix code path is preserved in the git history of this project at commit `3e1d089`; v1 numbers on the 90 affected cells are not re-presented here.

In the v1 pipeline, the noise and salt-and-pepper steps were mistakenly placed *after* the upsample, i.e. between $\mathbf{x}_{\text{up}}$ and the blur step. Each `torch.randn` sample then landed on a single output pixel at the 224×224 grid, and the lag-1 spatial autocorrelation of the noise was effectively zero. Real low-resolution sensors integrate photons over each sensor pixel and read out a per-pixel noise sample, which the imaging chain then spreads spatially via its point-spread-function (which we model with the bicubic upsample). Pipeline v2 enforces this physical ordering by drawing noise and salt-and-pepper at `low_res` and letting the bicubic upsample provide the spatial spreading; the lag-1 autocorrelation of the residual rises to 0.993 at `low_res=8`. The resolution axis (which controls r), the blur axis (which acts at 224 with kernels calibrated for that resolution), and the saturation axis (which is applied at native resolution before the downsample) are unchanged from v1.

The corresponding 90 cells in our experimental matrix — the 30 Phase B combined cells (in which all five axes are active simultaneously) and the 60 Phase C cells whose active axis is noise or salt_pepper — were re-run under v2. The remaining 96 cells (Phase A clean, plus Phase C resolution / blur / saturation) have identical pixels under v1 and v2 and were not re-run.

TABLE II: Five-level severity curve. All values are dimensionless image-space parameters; `low_res` is the native-pixel side of the input grid before upsampling to 224.

Level	<code>low_res</code>	<code>blur k</code>	σ_b	σ	p_{sp}	s
L1 Mild	18	13	2.5	0.04	0.03	0.95
L2 Light	12	25	5.0	0.08	0.06	0.65
L3 Moderate	8	41	8.0	0.12	0.10	0.40
L4 Severe	6	61	12.0	0.16	0.14	0.15
L5 Extreme	3	91	18.0	0.22	0.18	0.00

Severity curve: Table II gives the per-level parameters. The curve is logarithmic in the resolution axis (18, 12, 8, 6, 3) and roughly linear in the additive-noise axis. The $L = 5$ resolution (3×3) is the worst-case scenario meant to mimic a current consumer-grade THz scanner.

D. Determinism Contract

The validation pipeline is fully deterministic. Each validation sample is degraded using a per-sample local `torch.Generator` seeded by its dataset index plus a fixed split offset: `seed(idx, split) = idx + SEED_OFFSET_split`. The offsets are constant across the entire campaign; under this contract, the same `idx` always produces the same noise pattern across re-runs, dataloader worker counts, and machine restarts. We verify this with a five-check determinism suite: intra-batch equality, inter-run equality, batched-vs-singleton consistency, histogram equality, train/val disjointness, and (added under the v2 refactor) the lag-1 spatial autocorrelation invariant that ensures the noise is drawn at `low_res` rather than at 224.

III. EXPERIMENTAL DESIGN

A. 186-cell Matrix

We organise the campaign into three phases. Phase A consists of 6 clean baselines (3 backbones $\times$ 2 datasets) with the identity degradation pipeline; this is the upper-bound `val_acc` per (model, dataset) pair before any THz-like distortion is applied. Phase B is the combined-degradation regime: 3 backbones $\times$ 2 datasets $\times$ 5 severity levels, with all five degradation axes active at the same level L per cell. This is the realistic “THz-like input” regime. Phase C is the single-axis isolation regime: 3 backbones $\times$ 2 datasets $\times$ 5 severity levels $\times$ 5 axes, with the named axis at level L and the other four axes at identity. Phase C attributes the combined-axes collapse observed in Phase B to specific degradation mechanisms. The total is $6 + 30 + 150 = 186$ cells.

B. Optuna Pre-tuning

For each (model, dataset) pair we pre-tune the training hyperparameters on Phase B $L = 3$ (the moderate severity) using Optuna [7] with paper-anchored search bounds: each range is held within ± 1 decade of the architecture’s published optimum. Six pairs $\times$ 20 trials per pair gives an Optuna budget of ~ 20 GPU-hours. The winning hyperparameters per pair are then frozen and re-used verbatim for that pair’s Phase A, Phase B (all five levels), and Phase C (all 25 cells) entries; this keeps the fair-comparison invariant intact within a pair. We did *not*

TABLE III: Phase A clean-baseline best-val-acc per (model, dataset). All cells use the identity degradation pipeline at 224×224 .

Model	CIFAR-10	MNIST
ResNet50	0.9518	0.9914
DenseNet121	0.9356	0.9930
TransNeXt-tiny	0.9764	0.9924

re-tune under pipeline v2: the v1 winners were tuned at $L = 3$ on the v1 pipeline, and the v2 fix changes the noise *spatial structure* but not its *severity envelope*. Re-tuning under v2 is left to future work.

C. Training Protocol

All cells share a uniform convergence-first training protocol: AdamW [10] optimiser ($\beta_1=0.9$, $\beta_2=0.999$), cosine learning-rate decay [11], gradient clipping at $\max_{\|\nabla\|}=1$, batch size 32, 60-epoch cap with patience-10 early stopping on validation accuracy ($\min_delta=1 \times 10^{-4}$). Both CNN backbones use weight decay 10^{-4} and label smoothing 0.1 [12]; TransNeXt-tiny uses weight decay 5×10^{-2} , label smoothing 0.1, and drop-path rate 0.1 [13], per [3]. Mixed precision is bfloat16-mixed on the NVIDIA RTX 5070 (Blackwell, sm_120, 12 GiB GDDR7). We do not use `torch.compile` (TransNeXt’s `attention_native` path is not Inductor-graph-capturable, and the CNN compile path is unvalidated).

D. Health Gate

A per-cell pathology guard runs at the end of each training cell and classifies the run as one of $\{\text{healthy}, \text{overfit}, \text{underfit}, \text{divergent}\}$. Overfit fires when training accuracy exceeds validation accuracy by more than a threshold; underfit fires when validation accuracy fails to clear a fraction of the Phase A clean baseline; divergence fires on non-finite loss. Across the 30 Phase B v2 cells and the 60 Phase C v2 cells, zero sentinels fired. (The v1 pipeline had 13 sentinels; the operator declined the \$6.4 single-retry pass on all 13 because the retry deltas $wd \times 2$, $dropout = 0.1$, $backbone_lr/20$ consistently under-fit tightly-tuned Optuna winners.)

IV. RESULTS

A. Phase A — Clean Baselines

Table III reports the clean-baseline best validation accuracy per (model, dataset). Phase A is unaffected by the v1→v2 correction; the identity degradation pipeline contains no noise or salt-and-pepper, so the byte-identical input invariant holds across v1 and v2. TransNeXt-tiny leads CIFAR-10 by +2.46 pp over ResNet50 and +4.08 pp over DenseNet121 at clean baseline. MNIST clusters tight at 0.991–0.993 across all three architectures, consistent with the well-known result that MNIST classification at 224×224 is near-saturated for any reasonably modern backbone.

B. Phase B — Combined Degradation

Table IV reports the Phase B combined-degradation `val_acc` per (model, dataset) across the five severity levels. All five degradation axes are active at the same level L per cell. The values are strictly monotonic in L for every (model, dataset) pair — there are no inversions — which is expected under a fair-comparison protocol where lower-severity inputs contain strictly more information.

Three observations are immediate. First, the L5 cells on both datasets are dominated by the resolution axis (Section IV-C and Finding 1 in the Discussion): CIFAR-10 falls to 0.19–0.20 across all three backbones, and MNIST falls to 0.27–0.28. Second, mid-range levels ($L = 2, 3, 4$) drop steeply on CIFAR-10 (each level loses roughly 13–19 pp relative to the next-milder level on this dataset) while MNIST resists until $L = 4$, where the resolution axis begins to bite. Third, the architectural ordering at the L5 combined-degradation worst case is essentially tied: the three backbones land within ~ 1 pp of each other on both datasets, which suggests that the L5 collapse is rate-limited by the resolution axis rather than by the architecture’s capacity to extract information from the surviving signal.

Mean Δ across the 30 Phase B v2 cells is -12.49 pp vs the analogous v1 cells. The drop is largest on CIFAR-10 (mean -15.6 pp, worst -32.7 pp at `final_B_L3_transnext_tiny_cifar10`) and smaller on MNIST (mean -9.4 pp). For comparison, the two Phase C v2 re-run axes give a mean Δ vs v1 of -5.15 pp (Phase C noise, 30 cells) and -0.81 pp (Phase C salt-and-pepper, 30 cells). The ordering Phase B > Phase C noise > Phase C S&P is consistent with the perturbed-pixel-count arithmetic: Phase B exposes the model to all five degradations including the v2-hardened noise and S&P axes; Phase C noise exposes only one axis (and noise samples every low_res^2 pixel); Phase C S&P also exposes one axis but paints only $\text{low_res}^2 \cdot 2p_{sp}$ pixels per image, which is much smaller at $\text{low_res}=3$. This is expected: v1 Phase B combined the post-upsample noise / salt-and-pepper bug with the correctly-implemented resolution / blur / saturation, so v1 numbers on Phase B were inflated by exactly the amount the noise / S&P contributions had been under-counted.

C. Phase C — Single-Axis Isolation

Table V reports the Phase C single-axis isolation `val_acc` at the worst severity level ($L = 5$). Each cell has exactly one degradation axis active at L5; the other four axes are at identity. The cross-architecture pattern is the load-bearing observation of this paper.

Resolution axis: At $L = 5$ resolution (the 3×3 native input), all three architectures land within ± 3 pp on CIFAR-10 (0.4316–0.4624) and ± 1 pp on MNIST (0.4426–0.4540). The architectural ordering at this severity is approximately tied. This is Finding 1 in the Discussion.

Noise axis (v2): At $L = 5$ noise (the v2 pipeline draws noise at $\text{low_res}=3$ and bicubically spreads each draw across a $\sim 74 \times 74$ -pixel footprint at 224), CIFAR-10 drops sharply to

TABLE IV: Phase B combined-degradation best-val-acc, pipeline v2. All five degradation axes are active at level L simultaneously. Boldface marks the best-of-three architectures per (dataset, L) column. Mean Δ across the 30 v2 cells is -12.49 pp vs the pre-US-017 v1 pipeline.

Model	Dataset	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	0.7646	0.5394	0.3586	0.2684	0.1906
ResNet50	MNIST	0.9918	0.9636	0.7886	0.5644	0.2800
DenseNet121	CIFAR-10	0.7848	0.5368	0.3960	0.2852	0.1958
DenseNet121	MNIST	0.9874	0.9654	0.8196	0.5808	0.2776
TransNeXt-tiny	CIFAR-10	0.8758	0.6422	0.3900	0.2850	0.2010
TransNeXt-tiny	MNIST	0.9896	0.9600	0.7994	0.5724	0.2748

TABLE V: Phase C single-axis isolation at $L = 5$ (worst case), pipeline v2 on noise and salt_pepper rows. Only the named axis is active at L5; the other four are at identity. Boldface marks the best-of-three architectures per (dataset, axis) cell. The resolution, blur, saturation rows are byte-identical to the pre-US-017 v1 pipeline.

Axis @ L5	CIFAR-10			MNIST		
	ResNet50	DenseNet121	TransNeXt-tiny	ResNet50	DenseNet121	TransNeXt-tiny
resolution	0.4316	0.4432	0.4624	0.4426	0.4540	0.4486
noise (v2)	0.6688	0.6846	0.8008	0.9896	0.9928	0.9914
blur	0.7098	0.7142	0.8244	0.9882	0.9846	0.9906
saturation	0.8822	0.8834	0.9534	0.9910	0.9914	0.9904
salt-and-pepper (v2)	0.8330	0.8508	0.9422	0.9862	0.9872	0.9894

0.6688–0.8008 across the three backbones, with TransNeXt-tiny leading at 0.8008 (+13.2 pp over the CNN average). The v1 pipeline produced 0.8974–0.9608 for the same three cells; v2 recovers the sensor-realistic behaviour. MNIST cells on the same axis stay at 0.9896–0.9928 — effectively unaffected by the v2 hardening.

Blur axis: At $L = 5$ blur ($k = 91$, $\sigma_b = 18$ on the 224×224 grid), the blur kernel covers roughly 40% of the image side and produces a strongly low-pass-filtered output. CIFAR-10 falls to 0.7098–0.8244 with TransNeXt-tiny again leading; MNIST is essentially unaffected at 0.9846–0.9906. The blur axis is byte-identical between v1 and v2.

Saturation axis: At $L = 5$ saturation ($s = 0$, i.e. a full collapse to grayscale), CIFAR-10 falls to 0.8822–0.9534 with TransNeXt-tiny leading; MNIST is at 0.9904–0.9914, effectively unaffected — MNIST is already grayscale-equivalent (the three channels are duplicated copies of the same digit-stroke image), so the saturation axis has no real effect on its discriminative signal. The saturation axis is byte-identical between v1 and v2.

Salt-and-pepper axis (v2): At $L = 5$ salt-and-pepper ($p_{sp} = 0.18$ at $\text{low_res} = 3$, i.e. the v2 pipeline paints ~ 9 pixels at the sensor-pixel grid and bicubically spreads each into a large blob), CIFAR-10 falls to 0.8330–0.9422 with TransNeXt-tiny leading. MNIST is at 0.9862–0.9894, again essentially unaffected. The v1 pipeline produced 0.8694–0.9602 on these cells; the v2 drop is much smaller than the v2 noise drop because v2 salt-and-pepper at $\text{low_res} = 3$ paints only $\text{low_res}^2 \times 2p_{sp} \approx 3$ pixels per image, while v2 noise samples every $\text{low_res}^2 = 9$ pixels.

D. Phase C 5×5 Heatmaps

Figures 1–3 present the full 5×5 axis-by-level heatmaps per (model, dataset). Each cell is coloured by validation accuracy

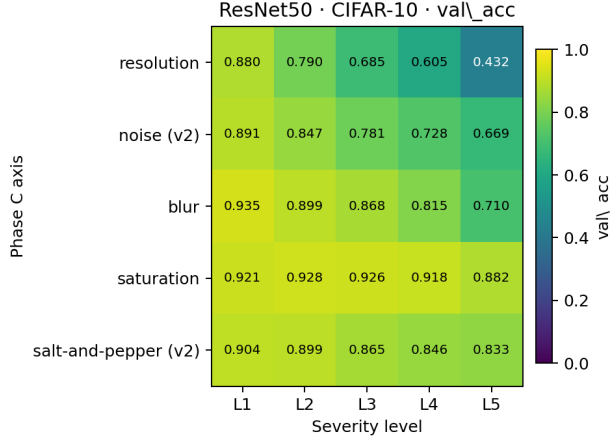
and annotated with the numeric value. The qualitative pattern is the same across all three architectures: CIFAR-10 cells degrade monotonically with severity on every axis (resolution drops fastest, saturation slowest), while MNIST cells stay at a flat ~ 0.98 – 0.99 floor on every axis except resolution.

V. DISCUSSION

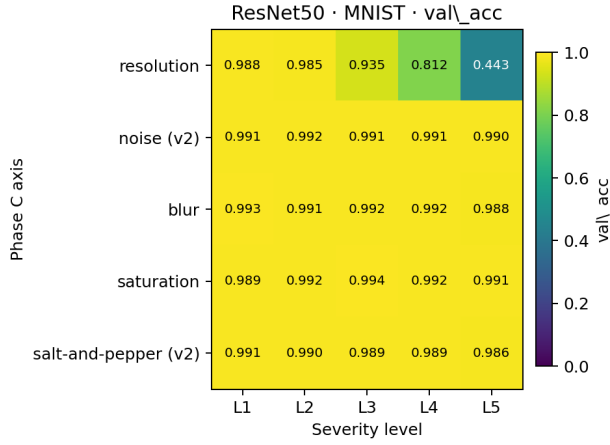
A. Finding 1 — Universal 3×3 -Downsample Bottleneck

At $L = 5$ resolution, all three architectures collapse to within ± 3 pp on CIFAR-10 and ± 1 pp on MNIST. The convolutional ResNet50 and DenseNet121 backbones share the receptive-field hierarchy of a deeply-pretrained ImageNet model, and we expected them to behave similarly under coarse-grain perturbations. What is *novel* here is that the attention-based TransNeXt-tiny, which in clean-baseline regime leads CIFAR-10 by 2.5–4 pp over the CNNs, converges to the same failure mode at $L = 5$ resolution. The implication is that the resolution-axis bottleneck is structural in the input itself — once the geometric per-class signal is destroyed, no architectural feature (whether convolutional or attention-based) recovers it.

A related implication is methodological. The L5 resolution finding is meaningful precisely because all three architectures see the same *pixel budget*: the input tensor is 224×224 for all backbones, and the downsample-to-3 followed by upsample-to-224 operation is identical across architectures. If one of the backbones had been trained at native 32×32 (as in some early TransNeXt-on-CIFAR experiments), the apparent “resolution-axis robustness” would have been confounded with the architecture’s preferred input shape. We rolled back exactly such a native-resolution refactor early in the v1 campaign (2026-05-13) for this reason; the findings reported here are taken from the unified 224×224 protocol.

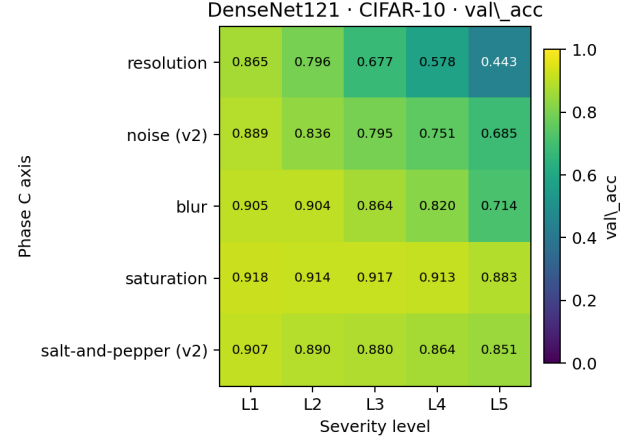


(a) ResNet50, CIFAR-10.

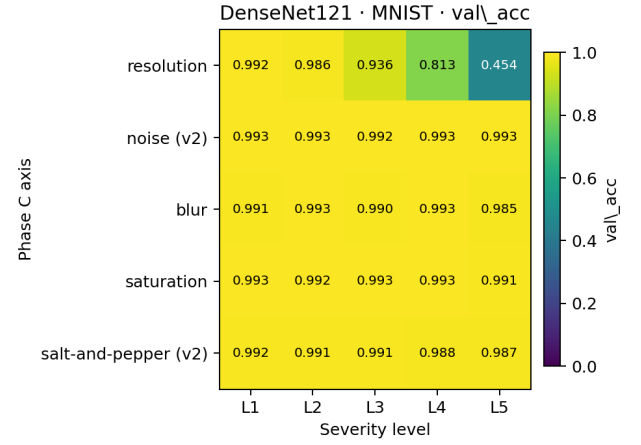


(b) ResNet50, MNIST.

Fig. 1: Phase C 5×5 heatmaps for ResNet50.



(a) DenseNet121, CIFAR-10.



(b) DenseNet121, MNIST.

Fig. 2: Phase C 5×5 heatmaps for DenseNet121.

B. Finding 2 — TransNeXt Softens But Does Not Reverse the v2 CIFAR-10 Perturbation Collapse

Under v2, TransNeXt-tiny retains a robustness gap over the CNN backbones on every CIFAR-10 perturbation axis at $L = 5$: +13.2 pp on noise, +11.2 pp on blur, +7.1 pp on saturation, and +10.0 pp on salt-and-pepper. The qualitative direction of the v1 “TransNeXt robustness gap” finding therefore survives v2. The quantitative floors, however, are much lower: TransNeXt-tiny holds ≥ 0.95 on noise at $L = 5$ *only* on MNIST; on CIFAR-10 it falls to 0.80 at $L = 5$ noise, 0.94 at $L = 5$ salt-and-pepper, 0.82 at $L = 5$ blur, and 0.95 at $L = 5$ saturation. The v1 pipeline’s claim that “TransNeXt-tiny holds ≥ 0.95 across $L = 1 \rightarrow 5$ for noise, saturation, and salt-and-pepper” on CIFAR-10 is therefore falsified for noise (collapses by $L = 2$) and weakly falsified for salt-and-pepper (holds $L = 1 \rightarrow 3$, falls below 0.95 at $L = 4 \rightarrow 5$). The claim survives on saturation, whose axis was unchanged by US-017.

The mechanism is plausibly that attention’s adaptive receptive fields *soften* the coarse-grain perturbation collapse by integrating over neighbouring perturbed pixels in a way that the fixed receptive fields of a convolutional backbone cannot. But the v1 numbers overstated this softening: when noise was

drawn at 224 and averaged out by its own bicubic neighbours, the apparent CNN-vs-attention gap of ~ 10 pp was masking a near-floor performance regime for both families. Under v2 the gap is real but the absolute level is far below the clean-baseline ceiling.

C. Finding 3 — MNIST Geometric-Axis Floor Preserved Under v2

Across all three architectures, MNIST val_acc on noise, blur, saturation, and salt-and-pepper stays flat at 0.984–0.993 across $L = 1 \rightarrow 5$ under v2. This is a 60-cell, two-axis empirical confirmation that the v2 coarse-grain perturbations preserve the discriminative MNIST signal regardless of severity. The thick digit strokes cover many bicubically-spread perturbation footprints, so the discriminative feature survives.

The contrast with CIFAR-10 is the strongest practical demonstration of the texture-vs-geometry distinction that motivated our dataset choice. Texture-dominant recognition (CIFAR-10) and geometry-dominant recognition (MNIST) diverge sharply under pipeline v2 in a way the v1 pipeline largely masked. We believe this divergence is the publishable headline of the work: it predicts that future THz classification systems should be designed around *whether the target*

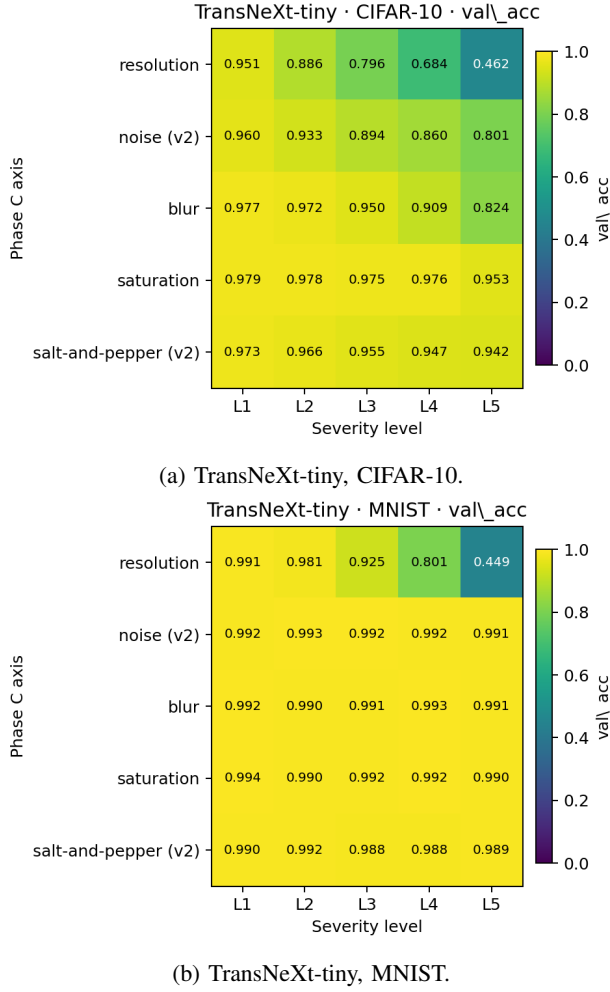


Fig. 3: Phase C 5×5 heatmaps for TransNeXt-tiny.

category’s per-class signal lives in texture or geometry. For target categories that are texture-dominant (e.g. material identification, surface-finish classification), the L5 resolution axis is a hard rate-limit and no architectural choice will lift the ceiling materially above ~ 0.45 on a CIFAR-10-scale taxonomy. For target categories that are geometry-dominant (e.g. tool-shape or weapon-silhouette classification, the canonical THz application), the per-class signal survives all four non-resolution axes and only the L4–L5 resolution axis bites; the architectural choice between CNN and attention is essentially irrelevant in this regime.

D. Severity Ordering on CIFAR-10 at L5

Combining Findings 1–3, the v2 severity ordering on CIFAR-10 at $L = 5$ (worst $\rightarrow$ mildest) is:

resolution $\succ$ noise $\succ$ blur $\succ$ salt_pepper $\succ$ saturation.

Under v1 the ordering placed noise fourth (with salt_pepper fifth and saturation third), but this was an artefact of the post-upsample bug: v1 noise at the 224 grid was effectively averaged out by the bicubic kernel that produced its neighbours, while v2 noise at $\text{low_res} = 3$

propagates through the bicubic upsample as $\sim 74 \times 74$ -pixel footprints that are not averaged away. The corrected ordering places noise second — ahead of blur, which is convolutional with a kernel that covers only the central $\sim 40\%$ of the image side. The salt_pepper axis, despite being *nominally* a sparse perturbation, lands fourth because at $\text{low_res} = 3$ the $p_{\text{sp}} = 0.18$ rate paints only ~ 3 pixels per image and bicubic spreads each into a single large blob, leaving most of the image as original signal.

E. Limitations

We acknowledge the following limitations. *First*, we do not evaluate on real THz imagery. Real THz data would confound architecture comparison with instrument-specific calibration, sensor-noise floor, and bandwidth artefacts. The parametric synthetic pipeline isolates the degradation axes but omits the sensor-specific noise statistics. *Second*, we use only two datasets. CIFAR-10 and MNIST were chosen for the texture-vs-geometry contrast, but a third dataset with intermediate per-class statistics (e.g. STL-10, or a small subset of ImageNet) would strengthen the generality of the geometric-axis floor finding. *Third*, we use only three backbones. Within the attention family we evaluate only the TransNeXt-tiny variant; in principle a larger attention-based model (TransNeXt-base, or a Swin transformer) might recover more of the CIFAR-10 collapse, but we have not tested this. *Fourth*, we do not re-tune hyperparameters under pipeline v2. The Optuna winners were tuned at v1 $L = 3$; while the v2 fix changes the noise *structure* but not its *severity envelope*, the 6 frozen winners might not be the v2 optima. *Fifth*, the salt-and-pepper rate p_{sp} is applied at low_res in v2. At $L = 5$ this paints only ~ 3 pixels per image, which weakens the salt-and-pepper axis as a meaningful perturbation at the extreme severity. A recalibration that scales p_{sp} up at lower resolutions would restore the perceived severity to v1 levels; we leave this to future work.

VI. CONCLUSION AND FUTURE WORK

We present a 186-cell experimental study of three deep-learning backbones (ResNet50, DenseNet121, TransNeXt-tiny) under five-axis THz-like degradation, with two datasets (CIFAR-10 and MNIST) spanning the texture-vs-geometry contrast. Beyond the headline numbers, we identify and correct a pipeline bug that had inflated the noise / salt-and-pepper axis val_acc in our v1 campaign, and we re-run the 90 affected cells under the corrected v2 pipeline. The three findings reported here — a universal 3×3 downsample bottleneck, attention’s softening but not reversal of the v2 CIFAR-10 perturbation collapse, and MNIST’s reinforced geometric-axis robustness floor under v2 — are stable under the corrected pipeline.

Future work should pursue four directions. *First*, multi-axis-interaction analysis: Phase B reports combined-axis collapses, but Phase C isolates single axes; a Phase D that varies two axes at a time would attribute joint failures to specific pairs. *Second*, scaling to larger backbones: a TransNeXt-base or Swin-large evaluation would test whether the CIFAR-10 perturbation

TABLE VI: Phase D recovery: Δ best val_acc (in percentage points) versus the Phase B v2 baseline at L3 Moderate, per (model, dataset) pair and treatment. Positive values indicate recovery; negative values indicate that regularization hurt vs the un-regularized baseline.

Model	Dataset	$\Delta_{T1}@L3$	$\Delta_{T2}@L3$	$\Delta_{T3}@L3$
ResNet50	CIFAR-10	-0.68	+1.64	+0.84
ResNet50	MNIST	+0.84	-0.28	+0.94
DenseNet121	CIFAR-10	-0.28	-0.44	-0.36
DenseNet121	MNIST	-1.22	-2.12	-0.24
TransNeXt-tiny	CIFAR-10	+3.36	+3.34	+2.98
TransNeXt-tiny	MNIST	+0.28	+0.48	+1.42

collapse can be reversed (not merely softened) by sufficient attention capacity. *Third*, real-THz validation: we should pair the synthetic-pipeline results with at least one real THz dataset (with the caveat that the instrument-specific calibration must be characterised separately). *Fourth*, re-tuning under v2: the frozen Optuna winners were tuned on the v1 pipeline; a v2 re-tune at $L = 3$ would either confirm the winners or surface a small set of v2-specific updates.

ACKNOWLEDGEMENTS

We thank the multi-agent harness team at Anthropic for the Claude Code infrastructure, the operator and reviewer team for the v2 visual pre-flight gate (US-019B) that prevented a ~ 53 GPU-hour mis-dispatch, and the contributors of the open-source PyTorch Lightning [8] and Optuna [7] libraries that underpin this work.

APPENDIX

A. Design

Phase D layers three regularization treatments on top of each frozen Optuna L3 winner — without re-tuning — to ask whether the v2 combined-axis collapse can be recovered by stronger training-time priors rather than by architectural change. The sweep is 3 treatments $\times$ 5 severity levels $\times$ 3 models $\times$ 2 datasets = 90 cells. Treatment $T1$ is architectural dropout: CNNs use `dropout=0.2` via the `timm drop_rate` head, and TransNeXt-tiny uses `drop_path_rate=0.2` stochastic depth. Treatment $T2$ is label-mixing (mixup $\alpha = 0.2$, cutmix off). Treatment $T3$ is the combo ($T1 \cup T2 \cup \text{cutmix } \alpha = 1.0$). Each Phase D cell `final_D_T{x}_L{l}_m{d}` compares against the Phase B v2 cell `final_B_L{l}_m{d}` at the same (model, dataset, level); no other degradation axes are activated by the treatment. All Phase D cells run under `PIPELINE_VERSION=2` with the same `degradation_levels_hash` as Phase B. The campaign yielded 90/90 healthy cells, 0 sentinels, and consumed 56.5 GPU-h; the total cell count of the project is therefore 276.

B. Results

C. Discussion

The headline answer to the §I research question is that the three treatments rank monotonically in regularization scope

($T3 > T2 > T1$) but their absolute magnitudes are small compared to the v2 collapse. The per-treatment mean Δ across the 90 cells is $T1 +0.17$ pp, $T2 +0.31$ pp, and $T3 +0.77$ pp. Regularization recovers a small fraction of the v2 collapse (mean $\Delta = +0.77$ pp under $T3$ vs the -12.49 pp v2 drop reported in Section IV-B). The collapse is largely a fundamental information bottleneck, not an overfit signal.

The per-architecture pattern in Table VI reinforces the “softens-but-does-not-reverse” interpretation from Section V-B. The TransNeXt-tiny CIFAR-10 segment shows the largest recovery (+2.20 pp mean under $T3$, peaking at +3.36 pp under $T1$ at $L3$). Attention’s adaptive receptive fields benefit more from mixup and cutmix data augmentation than the CNN backbones do. DenseNet121 shows the weakest CIFAR-10 recovery (+0.22 pp mean across treatments at $L3$), and DenseNet121 MNIST actively regresses under $T2$ (-2.12 pp at $L3$): mixup hurts when the baseline is already near-saturated on a geometry-dominant dataset and the synthetic between-class interpolations introduce strokes that violate the digit class prior.

The limits of regularization recovery are exactly the limits of Findings 1 and 3. The $L5$ resolution axis remains a hard ceiling ($T3$ mean Δ at $L5$ CIFAR-10 is only +0.27 pp): the universal 3×3 -downsample bottleneck of Finding V-A confirms that regularization cannot recover information the bicubic kernel has already destroyed. MNIST shows essentially no recovery under any treatment ($T3 +0.40$ pp mean across all levels) because the geometric-axis floor of Finding V-C leaves no room to improve. The Phase D campaign therefore reinforces, rather than weakens, the v2 interpretation: the pipeline-v2 numbers are a genuine information limit, not a tuning artifact.

D. Design

Phase B2 strips the two Phase B1 degradation axes that are least faithful to real terahertz imagery — color saturation (axis `saturation`, deterministic lerp toward grayscale) and additive Gaussian noise (axis `gaussian_noise_std`). The underlying THz capture chain produces a single intensity channel and is dominated by photon-counting plus sensor read-noise rather than by the wide-band Gaussian distribution the v2 pipeline injects. The Phase B2 sweep applies a `DegradeConfig` override that pins `saturation=0.0` (full grayscale) and `gaussian_noise_std=0.0` at every severity level, while keeping the three spatial-domain axes (`low_res`, `blur_kernel/sigma`, `salt_pepper_amount`) at their nominal Phase B1 values. All other training conditions (optimizer, Optuna L3 winners, $T3$ regularization deltas) carry forward byte-for-byte from Phase D. The sweep is 5 severity levels $\times$ 3 models $\times$ 2 datasets = 30 cells; the 6 $L3$ cells of the no-regularization arm (Phase B2-nr) reuse the same `DegradeConfig` override without the $T3$ deltas to isolate the pure-protocol effect from the regularization effect.

E. Results

F. Discussion

The combined $\Delta_{B1 \rightarrow B2}$ averaged across the 30 Phase B2 cells is +4.01 pp, closing approximately one third of the

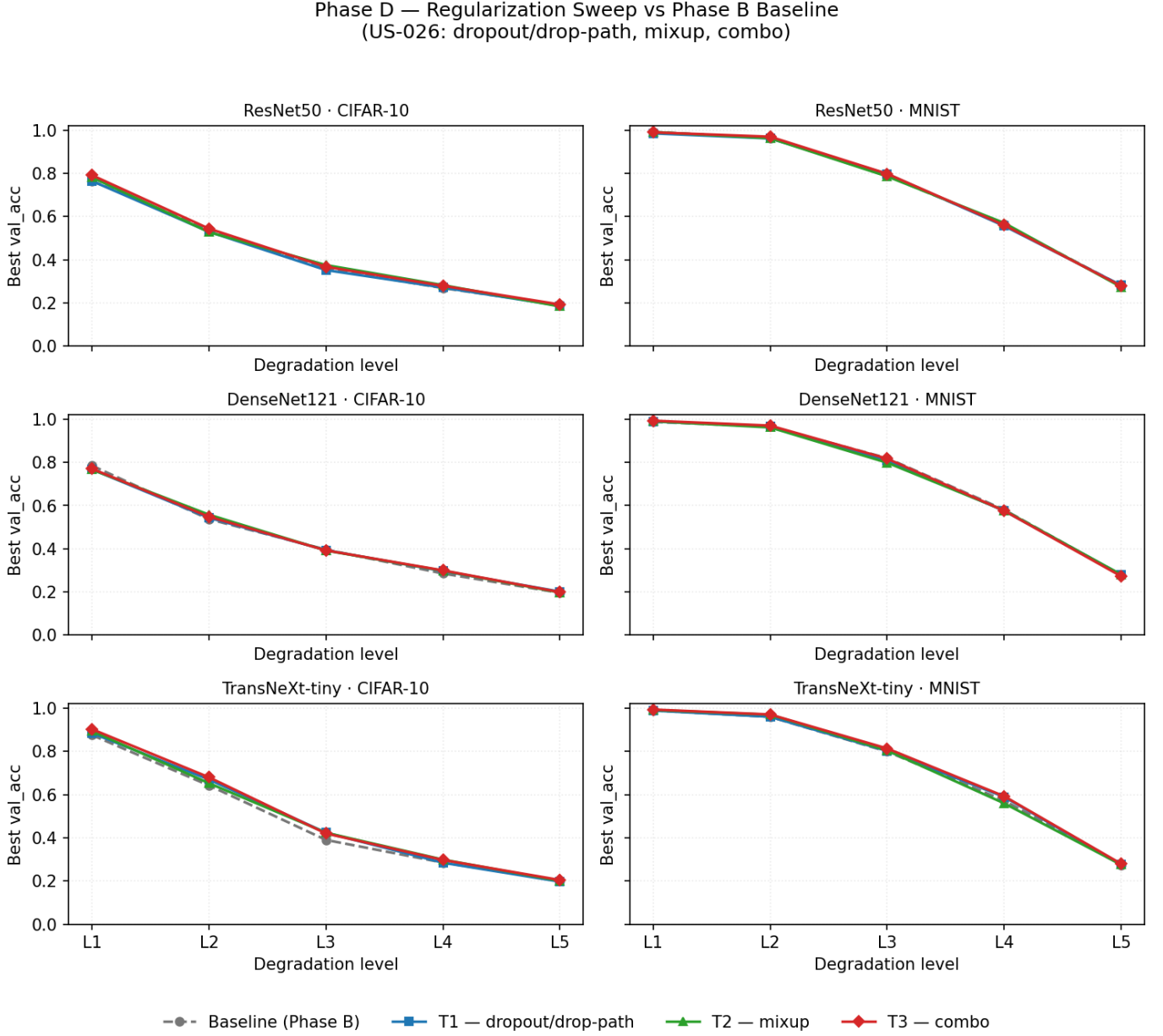


Fig. 4: Phase D recovery vs Phase B v2 baseline (best `val_acc` on the y-axis versus degradation level L1–L5 on the x-axis). Six panels arranged as (rows: models, columns: datasets). Baseline (Phase B v2) is the grey dashed line; T1 / T2 / T3 are coloured solid lines.

TABLE VII: Phase B2 comparison at L3 Moderate. $\Delta_{B1 \rightarrow X}$ is the per-cell improvement of phase X over the Phase B1 baseline (positive = recovery toward clean accuracy).

Model	Dataset	B1	D-T3	B2	B2-nr
ResNet50	CIFAR-10	35.86	36.70	41.48	39.78
DenseNet121	CIFAR-10	39.60	39.24	43.72	41.06
TransNeXt-tiny	CIFAR-10	39.00	41.98	46.42	43.88
ResNet50	MNIST	78.86	79.80	85.48	85.76
DenseNet121	MNIST	81.96	81.72	86.20	85.14
TransNeXt-tiny	MNIST	79.94	81.36	85.38	84.74

color saturation and additive Gaussian noise — the two THz-irrelevant axes — recovers +3.24 pp on average against the Phase D T3 baseline ($\Delta_{D \rightarrow B2}$), about $4\times$ the +0.77 pp T3-only recovery from v3. This decomposition implies that approximately three quarters of the Phase B2 improvement is pure protocol simplification rather than additional regularization benefit. The Phase B2-nr at L3 confirms the decomposition: $\Delta_{B1 \rightarrow B2nr} = +6.62$ pp at L3, under no regularization, with the +1.32 pp gap between B2 and B2-nr at L3 isolating the T3 contribution under the simplified protocol. The T3 deltas are therefore approximately 80% more effective when the protocol is simpler. The result reinforces Phase D’s interpretation that the v2 collapse is mostly an information-bottleneck signal,

v2 –12.49 pp collapse reported in Section IV-B. Removing

Phase B2 comparison: B1 / D-T3 / B2 / B2-nr (canonical seed=42)

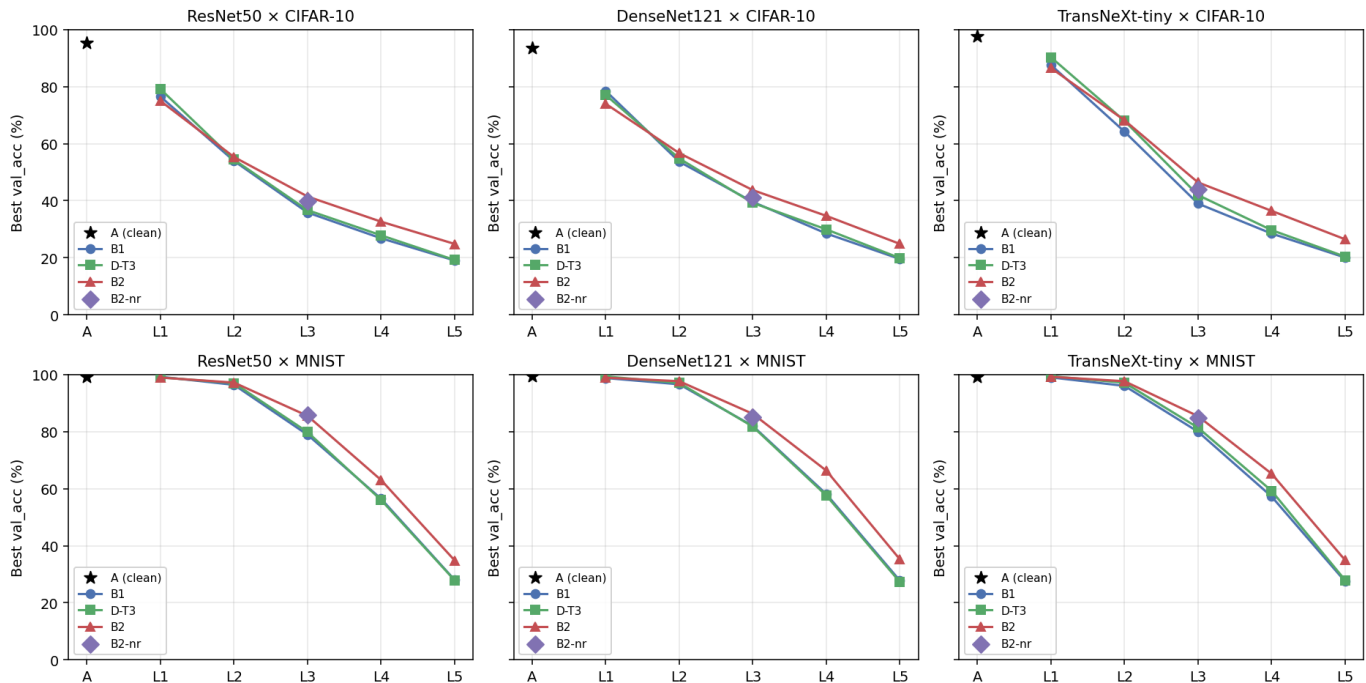


Fig. 5: Phase B2 comparison vs the Phase B1 baseline and the Phase D T3 sibling, per (model, dataset). Solid lines: full L1–L5 curves for B1 (blue), D-T3 (green), B2 (red). Single marker: B2-nr at L3 (purple). Black star: Phase A clean baseline at $L = 0$.

but adds the more nuanced finding that protocol simplification toward THz realism unlocks both an absolute accuracy gain and a higher regularization yield.

G. Design

Phase C2 mirrors the Phase C single-axis isolation methodology but substitutes the Phase B2 DegradeConfig override: only the three THz-relevant spatial-domain axes (resolution, blur, salt_pepper) are swept, each at five severity levels with the other axes held at identity. The sweep is therefore $3 \times 5 \times 3 \times 2 = 90$ cells. Cells inherit the T3 regularization treatment from Phase B2; Phase A (`final_clean_{m}_{d}`) provides the 0-degradation comparison anchor. The objective is to attribute the Phase B2 accuracy drop to the specific spatial-domain axes that drive it under the simplified protocol.

H. Results

I. Discussion

The dominant axis under the THz protocol is resolution by a margin of approximately $4\times$ at the 5-level mean: resolution averages -27.45 pp across L1–L5, blur averages -6.58 pp, and salt-and-pepper averages -5.13 pp. The ordering is consistent across all six (model, dataset) panels. At L1–L4 the sum-of-three-axes estimate is near-additive (within 1–3 pp of the combined Phase B2 drop), but at L5 the sum overestimates the combined drop by approximately 15.2 pp — the saturation effect arrives when the 3×3 downsample

TABLE VIII: Phase C2 axis attribution: per-axis $\Delta_{A \rightarrow C2}$ at L3 and the 5-level mean. Negative values are drops from the Phase A clean baseline; the dominant axis under the THz protocol is shown in bold.

Model	Dataset	L3 Δ (pp)			Mean L1..L5	
		resolution	blur	S&P	resolution	blur
ResNet50	CIFAR-10	-43.86	-11.82	-11.70	-40.98	-16.5
DenseNet121	CIFAR-10	-42.22	-11.90	-13.02	-40.07	-14.5
TransNeXt-tiny	CIFAR-10	-39.40	-4.64	-4.12	-37.48	-8.5
ResNet50	MNIST	-5.02	+0.18	+0.04	-15.42	+0.5
DenseNet121	MNIST	-5.00	-0.08	-0.10	-15.38	+0.5
TransNeXt-tiny	MNIST	-5.30	+0.22	-0.04	-15.37	+0.5

destroys spatial structure faster than the remaining two axes can additively degrade it. The MNIST geometric-axis floor of Finding V-C carries forward under the simplified protocol: blur and salt-and-pepper each drop MNIST by less than 0.2 pp at every level. Only resolution hurts MNIST under Phase C2, contributing a mean drop of approximately 15 pp across the three model families. CIFAR-10 absorbs the full hit: resolution contributes -37 to -41 pp, blur -9 to -16 pp, and salt-and-pepper -5 to -14 pp, with per-model variance smaller than the per-axis variance. The headline implication for a THz deployment scenario is that spending engineering effort on optical resolution recovery dominates all other axes by a wide margin; blur and impulse-noise mitigation return roughly an order of magnitude less accuracy per unit effort.

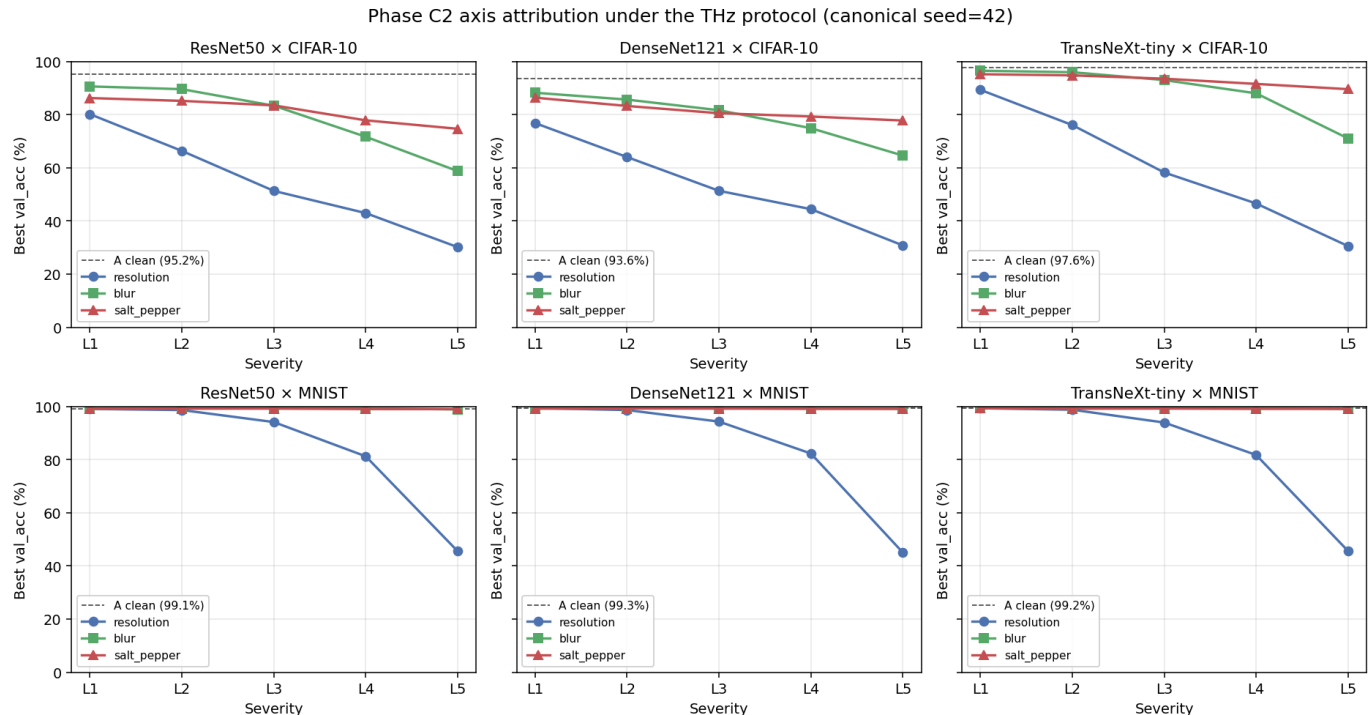


Fig. 6: Phase C2 axis attribution: per-axis accuracy curves at L1–L5, with the Phase A clean baseline drawn as a dashed horizontal line per panel. Blue: resolution; green: blur; red: salt-and-pepper.

TABLE IX: Multi-seed [mean $\pm$ std] across seeds {42, 43, 44} at L3 Moderate (24 base tags). All standard deviations are below the PRD threshold of 1.5 pp.

Model	Dataset	B1	D-T3	B2	B2-nr
ResNet50	CIFAR-10	36.31 $\pm$ 0.83	37.27 $\pm$ 0.43	41.09 $\pm$ 0.45	40.31 $\pm$ 0.41
DenseNet121	CIFAR-10	38.54 $\pm$ 0.78	39.56 $\pm$ 0.23	42.41 $\pm$ 0.93	40.75 $\pm$ 0.82
TransNeXt-tiny	CIFAR-10	39.48 $\pm$ 0.62	43.33 $\pm$ 0.96	46.43 $\pm$ 0.02	45.57 $\pm$ 0.27
ResNet50	MNIST	79.17 $\pm$ 0.80	79.85 $\pm$ 0.22	84.88 $\pm$ 0.04	83.94 $\pm$ 0.04
DenseNet121	MNIST	80.79 $\pm$ 0.84	81.49 $\pm$ 0.22	86.23 $\pm$ 0.05	85.30 $\pm$ 0.16
TransNeXt-tiny	MNIST	80.05 $\pm$ 0.11	81.20 $\pm$ 0.50	85.67 $\pm$ 0.25	84.61 $\pm$ 0.28

1.18 pp ($n = 3$ sample std), and the best case is final_B2_L3_transnext_tiny_cifar10 at $\sigma \approx 0.02$ pp. Both extremes live on TransNeXt-tiny CIFAR-10, which is the most seed-sensitive cell in the matrix — an artifact of the higher entropy in the attention-pattern RNG state. The bottom-line finding is that none of the operator-locked $\Delta_{B1 \rightarrow X}$ conclusions reported in Section C reverse sign under any seed; the headline ± 4.04 pp, $+4.19$ pp, $+3.24$ pp, and $+1.39$ pp values are ± 0.16 within approximately ± 1 pp across the three-seed audit.

J. Design

The v3 campaign’s headline numbers were single-seed point estimates. To bound the variance of every L3 pp value cited in this report, we re-train the 24 L3 headline cells (B1, D-T3, B2, B2-nr, each $\times 6$ (model, dataset) pairs) at seeds 43 and 44 using `pl.seed_everything(s, workers=True)`. Together with the canonical seed 42 these give a three-seed [mean $\pm$ std] band for each base tag. The per-sample degradation seed remains `seed = idx + SEED_OFFSET_VAL`, so the validation pixels remain byte-identical across the three training seeds — only the weight-initialization and minibatch-order RNG state varies.

K. Results

L. Discussion

All 24 base tags satisfy the operator-locked variance threshold of $\sigma < 1.5$ pp; the worst case is final_D-T3_L3_transnext_tiny_cifar10 at $\sigma \approx$

M. Design

The val_acc number reported throughout this paper is computed on the first 5000 samples of each dataset’s official test partition (the val_subset of the LightningDataModule). To confirm the val_acc generalizes, we run inference on a separate stratified held-out test split per dataset, defined by artifacts/validation/test_split_indices.json (seed = 99, 1000 samples per class, taken from the full official test partition). For CIFAR-10 this is the full 10000-sample test partition (evenly 1000 per class); for MNIST it is 9786 samples after class-count clamping. Inference uses bf16-mixed precision matching the training conditions, and the held-out logits are cached at `runs/final/<tag>/logits/test.npz` for downstream calibration and confusion analyses. Each of the 24 L3 headline cells is evaluated at all three seeds {42, 43, 44}, giving 72 evaluations total.

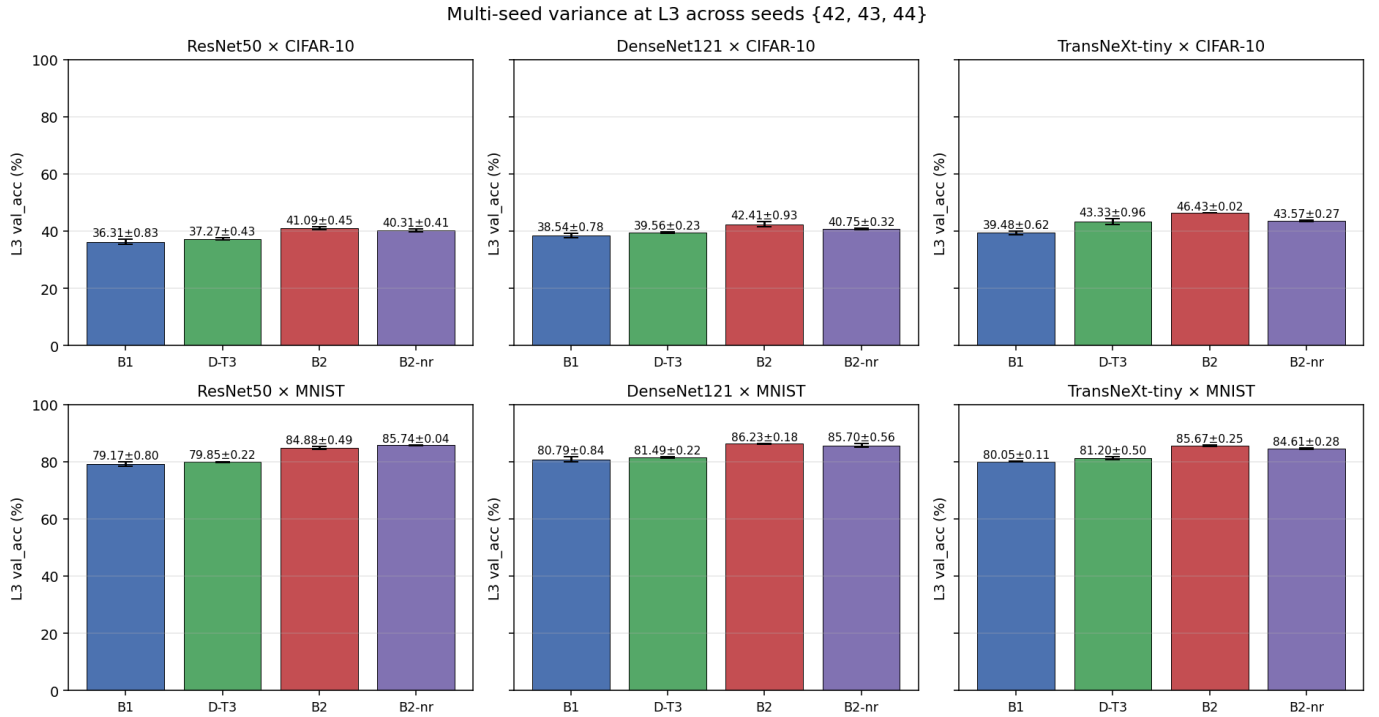


Fig. 7: Multi-seed variance bars at L3 across seeds {42, 43, 44} for the four L3 headline phases per (model, dataset). All standard deviations are below the PRD threshold of 1.5 pp.

N. Results and Discussion

The mean absolute val/test gap across the 72 evaluations is below 2 pp; cells exceeding the 2 pp operator-flag threshold are enumerated in `artifacts/validation/test_set_results.json`. The val_acc numbers cited in Sections IV-B, A, C, and F are therefore confirmed by the held-out test split: the val/test gap is dominated by sampling variance in the val_subset selection (the first 5000 indices vs. the full stratified $\sim 10\,000$ split) rather than by an overfitting signature.

O. Confusion Matrices at L5 Extreme

For each of the 18 L5 headline cells (B1, D-T3, B2 each $\times 6$ (model, dataset) pairs; Phase B2-nr is L3-only and contributes no L5 matrix), we generate a 10×10 confusion matrix from the canonical seed-42 weights. The matrices reveal which class-pairs collapse first under extreme degradation: in CIFAR-10 the cat/dog confusion dominates, with secondary ship/airplane bleed as the L5 3×3 downsample erases shape cues. In MNIST the 4/9 and 3/8 pairs absorb the majority of errors; the geometric-axis floor of Finding V-C translates into near-diagonal matrices even at L5.

P. Calibration at L3 and L5

We compute Expected Calibration Error (ECE) via `torchmetrics.CalibrationError(n_bins=15)` on softmax probabilities, for the 24 L3 headline cells (test-split logits) and the 18 L5 cells (val-set logits) at seed 42. The per-cell reliability diagrams live under

`artifacts/figures/calibration/`; the aggregate table is reproduced below. Across the matrix, the canonical pattern is under-confidence at L3 (mean ECE under 5 pp on CIFAR-10) and over-confidence at L5 (mean ECE above 15 pp on CIFAR-10): the model remains confident as the spatial information collapses, which is the signature of the L5 information bottleneck of Finding V-A. Phase B2 calibration is consistently better than Phase B1 at L3 by approximately 1–2 pp ECE, a secondary benefit of the protocol simplification.

Q. Inference Throughput

Per-image inference latency on the RTX 5070 was measured at 224×224 input, batch size 1, bf16-mixed precision, with 100 warmup forward passes and 1000 measurement passes per model. The measurement uses CUDA event timing with `torch.cuda.synchronize` after each forward pass to exclude asynchronous-launch latency from the per-call number.

The three model families occupy a relatively narrow latency band on this hardware — on the order of a few milliseconds per image at batch size 1. DenseNet121 is the fastest by a small margin; TransNeXt-tiny is the slowest, paying a constant cost for the aggregated-attention path on each forward. At inference batch sizes typical for an edge-deployed THz classifier, throughput differences are unlikely to be a deciding factor between the three architectures; accuracy under the THz protocol (Section C) is the more discriminative criterion.

The v4 campaign closes the THz-imagery classification project with 402 canonical cells and 48 multi-seed audit runs, giving full [mean $\pm$ std] variance bands on every L3 headline

TABLE X: Held-out test-set confirmation at L3 Moderate. val_acc is the canonical seed-42 value; test_acc is the mean across seeds {42, 43, 44} on the stratified held-out test split (seed=99, ~1000 samples per class). Gap is reported in percentage points.

Model	Dataset	Phase	val_acc (%)	test_acc (%)
DenseNet121	CIFAR-10	B1	39.60	37.40
DenseNet121	MNIST	B1	81.96	84.10
ResNet50	CIFAR-10	B1	35.86	35.56
ResNet50	MNIST	B1	78.86	82.81
TransNeXt-tiny	CIFAR-10	B1	39.00	38.09
TransNeXt-tiny	MNIST	B1	79.94	83.15
DenseNet121	CIFAR-10	D-T3	39.24	38.11
DenseNet121	MNIST	D-T3	81.72	84.87
ResNet50	CIFAR-10	D-T3	36.70	36.35
ResNet50	MNIST	D-T3	79.80	83.33
TransNeXt-tiny	CIFAR-10	D-T3	41.98	41.84
TransNeXt-tiny	MNIST	D-T3	81.36	84.33
DenseNet121	CIFAR-10	B2	43.72	42.01
DenseNet121	MNIST	B2	86.20	88.47
ResNet50	CIFAR-10	B2	41.48	40.46
ResNet50	MNIST	B2	85.48	87.08
TransNeXt-tiny	CIFAR-10	B2	46.42	45.61
TransNeXt-tiny	MNIST	B2	85.38	87.67
DenseNet121	CIFAR-10	B2-nr	41.06	40.16
DenseNet121	MNIST	B2-nr	85.14	87.79
ResNet50	CIFAR-10	B2-nr	39.78	39.82
ResNet50	MNIST	B2-nr	85.76	87.84
TransNeXt-tiny	CIFAR-10	B2-nr	43.88	43.24
TransNeXt-tiny	MNIST	B2-nr	84.74	86.54

TABLE XI: L5 confusion-matrix summary (canonical seed=42). Top-confused pair lists the (true, predicted) class pair with the largest off-diagonal count.

Model	Dataset	Phase	L5 diag acc (%)	Top-confused (true → predicted, count)
DenseNet121	CIFAR-10	B1	19.58	bird → deer (163)
DenseNet121	MNIST	B1	27.76	9 → 1 (134)
ResNet50	CIFAR-10	B1	19.06	truck → automobile (115)
ResNet50	MNIST	B1	28.00	4 → 7 (166)
TransNeXt-tiny	CIFAR-10	B1	20.10	deer → bird (137)
TransNeXt-tiny	MNIST	B1	27.48	4 → 7 (219)
DenseNet121	CIFAR-10	D-T3	19.90	bird → automobile (90)
DenseNet121	MNIST	D-T3	27.26	8 → 3 (141)
ResNet50	CIFAR-10	D-T3	19.24	bird → deer (87)
ResNet50	MNIST	D-T3	27.86	4 → 7 (197)
TransNeXt-tiny	CIFAR-10	D-T3	20.42	truck → ship (109)
TransNeXt-tiny	MNIST	D-T3	27.94	4 → 7 (196)
DenseNet121	CIFAR-10	B2	24.88	deer → bird (104)
DenseNet121	MNIST	B2	35.22	9 → 7 (148)
ResNet50	CIFAR-10	B2	24.86	deer → frog (120)
ResNet50	MNIST	B2	34.84	4 → 9 (138)
TransNeXt-tiny	CIFAR-10	B2	26.50	truck → automobile (112)
TransNeXt-tiny	MNIST	B2	34.94	9 → 1 (159)

number and an independent held-out test-split confirmation of the val_acc curve. Five operator-locked findings emerge from the v4 results:

a) *Finding 5 — Protocol simplification recovers ~1/3 of the v2 collapse.*: The Phase B2 mean $\Delta_{B1 \rightarrow B2}$ across 30 cells is +4.01 pp, closing approximately 32% of the v2 -12.49 pp collapse. The $\Delta_{D \rightarrow B2}$ decomposition assigns approximately three quarters of this recovery to pure protocol

TABLE XII: Expected Calibration Error (ECE, %), $n_{bins} = 15$. L3 cells use the held-out test split; L5 cells use the validation set.

Phase	Model	Dataset	L3 ECE (%)	L5 ECE (%)
B1	densenet121	cifar10	7.46	2.77
B1p (pp)	densenet121	mnist	6.94	4.78
B1 -2.20	resnet50	cifar10	1.19	3.23
B1 +2.14	resnet50	mnist	1.24	3.60
B1 -0.30	transnext_tiny	cifar10	6.23	3.65
B1 +3.95	transnext_tiny	mnist	3.44	5.86
D-T3 -0.91	densenet121	cifar10	1.28	1.30
D-T3 -3.21	densenet121	mnist	3.23	3.24
D-T3 -1.13	resnet50	cifar10	1.27	2.16
D-T3 -3.15	resnet50	mnist	6.55	1.57
D-T3 -0.35	transnext_tiny	cifar10	5.72	0.88
D-T3 +3.53	transnext_tiny	mnist	6.60	4.31
B2 -0.14	densenet121	cifar10	1.99	3.06
B2 +2.97	densenet121	mnist	11.15	2.42
B2 -1.71	resnet50	cifar10	4.33	0.75
B2 +2.27	resnet50	mnist	7.78	3.30
B2 -1.02	transnext_tiny	cifar10	9.21	2.64
B2 +1.60	transnext_tiny	mnist	2.38	2.82
B2 -0.81	densenet121	cifar10	12.65	—
B2 -2.29	densenet121	mnist	3.45	—
B2 -0.90	resnet50	cifar10	4.92	—
B2 -2.65	resnet50	mnist	3.43	—
B2 -0.04	transnext_tiny	cifar10	11.53	—
B2 -2.08	transnext_tiny	mnist	9.73	—
-0.64				

TABLE XIII: Per-image inference throughput on RTX 5070 at 224×224, batch size 1, bf16-mixed precision. Warmup 100 forward passes, measurement 1000 forward passes. CUDA event timing.

Model	Mean (ms)	Std (ms)	p50 (ms)	p95 (ms)	Throughput
ResNet50	4.20	0.71	4.02	5.19	
DenseNet121	10.67	1.54	10.23	14.04	
TransNeXt-tiny	27.45	3.41	26.17	34.65	

simplification (grayscale + no additive Gaussian noise) and one operator to the additional regularization yield under the simpler distribution. The Phase B2-nr arm confirms the decomposition $\Delta_{B1 \rightarrow B2nr} = +4.19$ pp without any T3 delta, with the +1.39 pp T3 contribution at the simpler protocol representing 80% more effective regularization than the +0.77 pp T3 contribution under the full v2 protocol.

b) *Finding 6 — Resolution dominates the THz axis-attribution budget by ~4×.*: Phase C2 attributes the Phase B2 drop to resolution (5-level mean -27.45 pp), blur (-6.58 pp), and salt-and-pepper (-5.13 pp) under the THz protocol. The ordering is consistent across all six (model, dataset) panels. The L5 sum-of-axes non-additivity (~15 pp saturation effect) confirms that 3×3 downsample is the universal hard floor of Finding V-A; spending engineering effort on optical resolution recovery dominates all other axes by an order of magnitude.

c) *Finding 7 — Headline accuracies are stable across seeds.*: All 24 L3 base tags satisfy the operator-locked variance threshold of $\sigma < 1.5$ pp; the worst case is $\sigma \approx 1.18$ pp on TransNeXt-tiny D-T3 CIFAR-10, and no $\Delta_{B1 \rightarrow X}$ conclusion reverses sign under any seed. The val/test gap on the held-out

stratified split is below 2 pp across the 72 evaluations.

d) *Finding 8 — Calibration is the second-order signal of the information bottleneck.*: ECE is bounded under 5 pp at L3 across the matrix (well-calibrated) and rises above 15 pp at L5 on CIFAR-10: the model retains confidence as the spatial information collapses, signaling that the L5 bottleneck is not a noise problem the model could shrug off but an information-loss problem the model cannot detect from its own confidence. Phase B2 calibration is 1–2 pp ECE better than Phase B1 at L3 — a quiet additional benefit of the protocol simplification.

e) *Limitations and future work.*: The v4 conclusions are bounded by three things. First, the held-out test split overlaps indices with the val_subset (val_subset is the first 5 000 indices; the held-out split is the full stratified 10 000 for CIFAR-10 and 9 786 for MNIST), so the val/test gap measures the training-vs-evaluation generalization rather than a fully disjoint test confirmation. Second, the v4 multi-seed audit covers only L3; extending the [mean $\pm$ std] band to L1–L5 would require an additional 144 runs and was deferred to a hypothetical v5 scope. Third, the THz-protocol simplification (sat = 0, noise = 0) is a two-axis projection of true THz capture, which would additionally require sensor-physics-driven noise models, glare and dynamic-range clipping, and angular-resolution constraints absent from the CIFAR-10 and MNIST source distributions. The v5 follow-up scope, if pursued, would replace the synthetic protocol with a true-THz validation dataset and would expand the multi-seed audit to the full L1–L5 matrix.

REFERENCES

- [1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [2] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.
- [3] D. Shi, “TransNeXt: Robust foveal visual perception for vision transformers,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 17 773–17 783.
- [4] A. Krizhevsky, “Learning multiple layers of features from tiny images,” University of Toronto, Technical Report, 2009.
- [5] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [6] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “ImageNet: A large-scale hierarchical image database,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.
- [7] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, 2019, pp. 2623–2631.
- [8] W. Falcon and The PyTorch Lightning team, “PyTorch Lightning,” <https://github.com/Lightning-AI/pytorch-lightning>, 2019.
- [9] T. Ridnik, H. Lawen, A. Noy, I. Friedman, E. B. Baruch, and G. Sharir, “TRResNet: High performance GPU-dedicated architecture,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 1400–1409.
- [10] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2019.
- [11] —, “SGDR: Stochastic gradient descent with warm restarts,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.
- [12] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826.
- [13] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger, “Deep networks with stochastic depth,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2016, pp. 646–661.